

S4G Image Workflow Documentation

Overview

The workflow downloads S4G FITS images, builds a geometry manifest, and plots diagnostic isophote PDFs.

Workflow:

1. download_s4g_images.py

Downloads each galaxy's S4G 3.6 micron FITS image from IRSA.

Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

2. build_s4g_geometry_manifest.py

Combines the downloaded FITS image paths with local and catalogue geometry data.

Output: Erwin_s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv

3. plot_s4g_isophote_axes.py

Reads the manifest and FITS images, then creates major/minor-axis isophote diagnostic PDFs.

Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Function-level overview: download_s4g_images.py

- parse_args: Defines command-line options such as --limit, --dry-run, --outdir, and --timeout.
- read_galaxy_names: Reads scrambled_map.txt and returns the galaxy names from the third column.
- download_image: Builds the IRSA FITS URL, skips existing files, optionally dry-runs, and saves valid downloads.

- `main`: Resolves paths, creates the output directory, reads the galaxy list, and downloads each requested image.

Function-level overview: `build_s4g_geometry_manifest.py`

- `clean_value`: Converts table text to numbers and normalizes known missing or invalid values to `None`.
- `read_scrambled_map`: Loads scrambled index, original index, and galaxy name from `scrambled_map.txt`.
- `read_s4g_table`: Reads local `s4gbars_table.dat` and keeps cleaned S4G bar and galaxy fields by name.
- `table_value`: Converts Astropy masked, scalar, or byte values into ordinary Python values.
- `read_cached_or_fetch_table`: Loads a VizieR catalogue from cache or downloads it and writes an ECSV cache file.
- `keyed_table`: Turns a catalogue table into a dictionary keyed by a selected name column.
- `keyed_herrera_bars`: Filters Herrera-Endoqui rows to bar entries and keys them by galaxy name.
- `fetch_catalog_geometry`: Fetches or loads Herrera-Endoqui, Salo, and Diaz-Garcia geometry catalogues.
- `fits_metadata`: Reads FITS header metadata needed for image size, pixel scale, centre, and orientation.
- `build_manifest`: Creates one manifest row per galaxy, merging local tables, catalogue geometry, FITS metadata, and notes.
- `write_manifest`: Writes the manifest rows to CSV using the fixed `OUTPUT_FIELDS` column order.
- `parse_args`: Defines paths and catalogue-cache switches such as `--image-dir`, `--no-vizier`, and `--refresh-cache`.

- main: Loads optional catalogue geometry, builds the manifest, writes the CSV, and prints image-link counts.

Function-level overview: `plot_s4g_isophote_axes.py`

- `parse_float`: Safely converts manifest string values to finite floats.
- `safe_filename`: Creates filesystem-safe PDF names from galaxy names.
- `read_manifest`: Reads the geometry manifest CSV into dictionaries.
- `pa_endpoint`: Converts a position angle and radius into x/y endpoints for drawing or sampling lines.
- `draw_pa_line`: Draws major or minor axis lines on the image panel.
- `extract_centered_subimage`: Crops a centre-based image cutout and builds arcsecond coordinate arrays.
- `profile_at_pa`: Samples a smoothed intensity profile along a requested position angle.
- `robust_log_image`: Creates a stable log-scaled image and contour levels even when pixels include zeros or NaNs.
- `deprojected_profile_radius`: Uses `angle_utils.deprojectr` to convert projected profile radii to deprojected radii.
- `required_geometry`: Pulls required geometry fields from a manifest row and rejects incomplete rows.
- `make_plot`: Creates the two-panel diagnostic plot for a single galaxy and writes it to individual and/or combined PDFs.
- `selected_rows`: Applies `--names` and `--limit` filters to the manifest rows.
- `parse_args`: Defines plotting options such as `--manifest`, `--image-dir`, `--output-dir`, `--names`, and `--no-combined`.
- main: Creates output folders, iterates through selected galaxies, writes PDFs, and reports skipped cases.

Program detail: download_s4g_images.py

Purpose: downloads S4G 3.6 micron FITS images from IRSA for galaxies listed in scrambled_map.txt.

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

Inputs: the scrambled map file supplies the galaxy names from its third column.

Behaviour: existing FITS files are skipped, --dry-run prints planned downloads, and --limit supports small test batches.

Key options: --galaxy-list, --outdir, --base-url, --limit, --timeout, and --dry-run.

Key commands: python download_s4g_images.py --dry-run --limit 5; python download_s4g_images.py --limit 1; python download_s4g_images.py

Program detail: build_s4g_geometry_manifest.py

Purpose: builds a CSV manifest that links each galaxy to FITS metadata and geometry measurements.

Default output: Erwin_s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv

Default image folder: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

Default cache folder: Erwin_s4g_image_downloader\geometry_catalog_cache.

Inputs: scrambled_map.txt, s4gbars_table.dat, downloaded FITS headers, and optional VizieR catalogues.

Catalogue sources: Herrera-Endoqui+2015, Salo+2015, and Diaz-Garcia+2016.

Output fields include image_exists, image_path, pixel scale, CRPIX/CRVAL, disc geometry, bar geometry, stellar mass, and notes.

Key options: --scrambled-map, --s4g-table, --image-dir, --cache-dir, --no-vizier, --refresh-cache, and --output-csv.

Key commands: python build_s4g_geometry_manifest.py; python build_s4g_geometry_manifest.py --no-

vizier; python build_s4g_geometry_manifest.py --refresh-cache

Program detail: plot_s4g_isophote_axes.py

Purpose: creates Figure-1-style diagnostic PDFs for each selected galaxy.

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Default manifest: Erwin_s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv.

Default image fallback folder: D:\Dropbox\Public Documents\UCLAN\MSc

Research\Erwin\s4g_images_36um.

Inputs: the geometry manifest and the downloaded FITS images.

Plot contents: log-scaled S4G isophotes, observed bar major axis, projected minor axis, and major/minor-axis intensity cuts.

Behaviour: incomplete rows or missing FITS images are skipped and reported after the batch.

Key options: --manifest, --image-dir, --output-dir, --combined-pdf, --limit, --names, --profile-width, --no-individual, and --no-combined.

Key commands: python plot_s4g_isophote_axes.py; python plot_s4g_isophote_axes.py --limit 3; python plot_s4g_isophote_axes.py --names NGC1879 IC0600

Recommended run order:

1. python download_s4g_images.py --dry-run --limit 5
2. python download_s4g_images.py --limit 5
3. python build_s4g_geometry_manifest.py --no-vizier
4. python plot_s4g_isophote_axes.py --limit 3